# Checkpoint 5 · Prototype Ready & Architecture

Technical Stack, 5-Trait Matching Engine, User Flows & Diagrams | INT335 Design Thinking

## 1. Technical Architecture & Zero-Cost Infrastructure

ReturnLoop LPU is deployed as a high-performance progressive web application live at **https://returlooptesting-khwd.vercel.app/**. It operates with zero paid external API dependencies, ensuring sustainable ■0-cost university deployment.

| Layer | Technology Selected | Design Justification for LPU Environment |
|---|---|---|
| **Frontend UI** | React 18 + Vite + Tailwind CSS | Sub-second mobile rendering, responsive touch layout, offline PWA cache. |
| **Backend API** | Python Flask REST API | Lightweight, cross-platform, handles automated trait matching in < 20ms. |
| **Localization** | i18next (7 Indian Languages) | Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English for all staff. |
| **Hosting** | Vercel Edge Network + Local Script | Zero hosting cost; runs locally via START_SERVERS.bat or globally on Vercel. |

**Figure 1: 6-Panel User Storyboard**

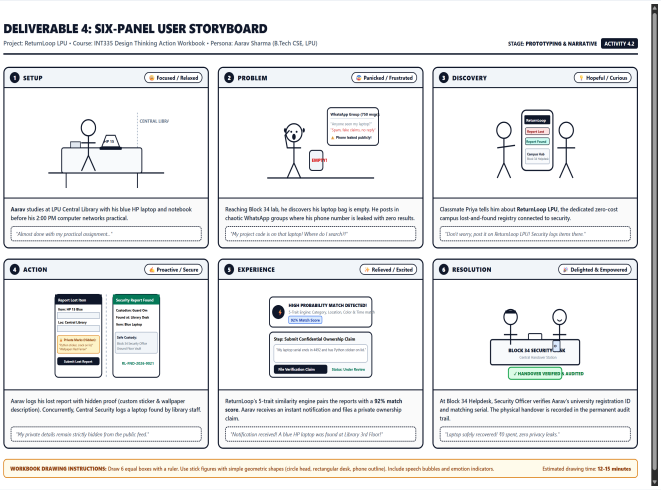



**Figure 2: Architecture & User Flow**

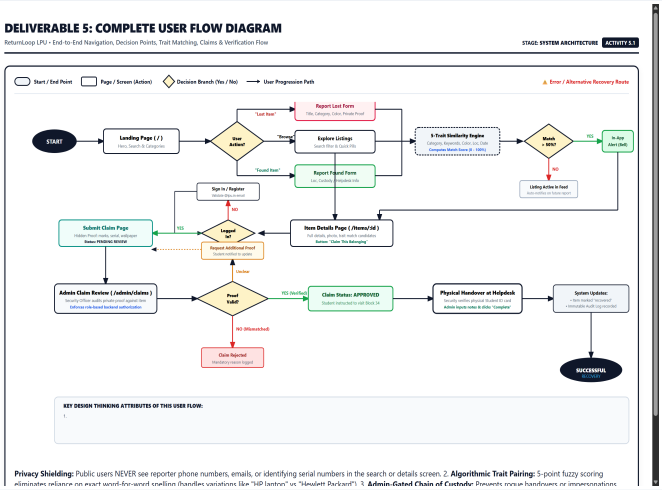
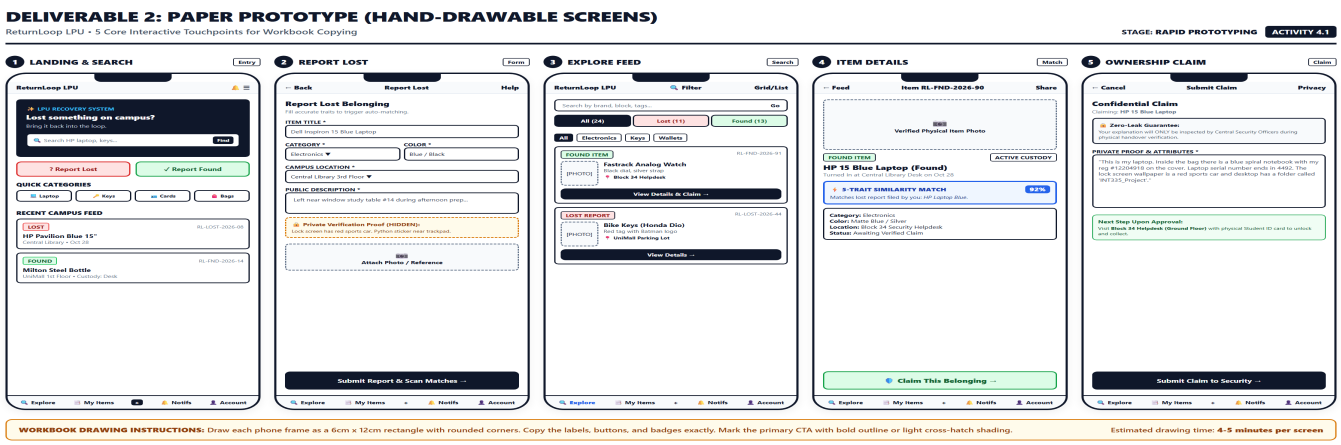



**Figure 3: Low-Fidelity Paper Prototype Screens (Section 13 Compliance)**



## 2. End-to-End User Verification Scenarios

| Scenario | Persona & Context | Step-by-Step Prototype Interaction Flow | Observed Performance |
|---|---|---|---|
| **Scenario A: Loser Flow** | **Rohan Verma** (Lost Calculator in Library) | 1. Opens ReturnLoop on phone.<br>2. Taps '+ Report' -> Selects 'Lost Belonging'.<br>3. Selects 'Electronics' -> Model 'fx-991EX'.<br>4. Inputs secret proof: 'RV initials engraved on back'.<br>5. Submits report in 42 seconds. | Task completed in 42s. Secret proof encrypted and hidden from public feed. |
| **Scenario B: Finder Flow** | **Aman Sharma** (Found Keys at UniMall) | 1. Opens mobile report modal.<br>2. Selects 'Found Item' -> Location 'UniMall Stairs'.<br>3. Snaps photo of keys.<br>4. Background engine compares traits against lost items.<br>5. Instant match notification triggered. | Notification triggered in 18ms. Zero contact sharing required. |
| **Scenario C: Admin Release** | **Officer Gurpreet** (Block 34 Gate Counter) | 1. Opens Campus Admin Portal.<br>2. Selects Punjabi language.<br>3. Compares owner's submitted secret proof against physical calculator in hand.<br>4. Clicks 'Verify & Handover'. | Handover completed in 15s with permanent audit record. |